

Spring Boot 4

What's New and Why It Matters



This isn't a minor update.

Spring Boot 4 is a major evolution

built for:

- **Modern Java (Java 21)**
- **Kubernetes-native apps**
- **Cloud scalability**
- **Lightning-fast startup ⚡**



- **Java 21 + Virtual Threads**
- **Handle thousands of concurrent requests effortlessly**
- **Lightweight threads = no memory bloat**
- **Perfect for high-performance REST APIs**



- **Faster Starts with AOT Compilation**
- **AOT (Ahead-of-Time) processing just got better.**
- **Optimized for GraalVM native images**



- **Smarter Dependency Management**
- **Auto-resolves compatible library versions**
- **Aligns with Spring Framework 6.1**
- **Less manual exclusions & version conflicts**



- **Observability-First Approach**
- **Unified metrics, logs, and traces**
- **Built-in support for Prometheus, Grafana, OpenTelemetry**
- **New health, liveness & readiness probes**



- **Cloud and Kubernetes Ready**
- **ConfigMap & Secret binding**
- **Built-in service discovery**
- **Ready for containerized and serverless deployments**



- **Enhanced support for HTTP interfaces (declarative REST clients).**
- **Better integration with Spring HATEOAS and JSON:API standards.**
- **Simplified exception handling with structured error responses.**
- **Example:**
`@HttpExchange("/api/users")
public interface UserClient {
 @GetExchange("/{id}")
 User getUserById(@PathVariable Long id);}`



- **3. Migration: How to Upgrade from Spring Boot 3**
- **Upgrading is straightforward if you're already on Spring Boot 3.x**
- **Update your parent dependency in pom.xml or build.gradle**
- **Ensure your Java version is 21 or later**
- **Remove deprecated properties (spring.main.lazy-initialization, etc.)**
- **Review Actuator endpoints; a few have been renamed**
- **Test thoroughly with spring-boot-starter-test (JUnit 5 support continues).**



- **Declarative HTTP security**
- **OAuth2 & JWT-ready**
- **Built-in CORS + password encoding**